

A Train-Free Pilot Diagnostic for Gain/Loss Framing Effects in Large Language Models

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Abstract

Gain/loss framing studies ask whether equivalent decisions elicit different behavior when outcomes are described as gains or losses, but LLM evaluations often lack paired controls that separate true frame robustness from preference erasure or lexical shortcuts. We introduce a train-free paired-source diagnostic that keeps 25 source items from four retained families fixed while generating gain/loss, neutral, counterframe, and lexical-control variants for the same hosted chat backend. The evaluation compares raw prompting, reframing and self-checking baselines, a canonical decision-normalization candidate, preference triangulation, ablations, and a deterministic lexical diagnostic under one row-complete matrix. The 1,260-row run has zero parse failures and a negative primary result: canonical decision normalization scores 0.565 macro mean versus 0.574 for raw prompting, barely changes frame-sensitivity choice, and has paired bootstrap intervals against raw that include zero, while preference triangulation becomes only a train-free reference floor and the lexical diagnostic reaches 0.940 macro mean score. The contribution is therefore a calibrated diagnostic recipe and negative comparator result: future gain/loss framing claims should pair source-level accounting, preference preservation, uncertainty, lexical controls, and payoff extraction before reporting mechanism or mitigation conclusions.

Introduction

Gain/loss framing is a classic case where an equivalent decision can elicit different risk preferences depending on whether outcomes are described as saved lives, lost lives, gains, or losses (Kahneman and Tversky 1979; Tversky and Kahneman 1981). In language-model systems, the analogous failure mode is practical rather than only psychological. A prompt, evaluator rubric, or source description may change a model’s recommended action even when the underlying decision target is unchanged. Recent studies therefore treat cognitive bias, source framing, evaluator framing, and prospect-theory-like behavior as measurable LLM behaviors (Suri et al. 2023; Echterhoff et al. 2024; Koo et al. 2023; Germani and Spitale 2025; Wang and Zhang 2026; Payne 2025; Wang et al. 2025).

The evaluation problem is that a higher aggregate score does not by itself show mitigation. Cognitive-bias suites and evaluator-bias benchmarks show that model behavior can change across formulations (Echterhoff et al. 2024; Koo

et al. 2023; Lyu et al. 2025). Train-free self-correction or re-framing can also alter an answer distribution without proving that the original preference was preserved. A gain/loss framing evaluation therefore needs paired frames, source-level accounting, preference-preservation diagnostics, and lexical controls that ask whether the signal is recoverable from surface words.

This gap matters because train-free methods are easy to deploy and easy to overread. A prompt wrapper can improve a metric by changing refusal style, washing out the original preference, or exploiting the terms that appear in a transformed source item. The evaluation target in this paper is therefore not an unconstrained score increase. It is the narrower question of whether a candidate preserves the decision target while reducing frame sensitivity under matched source items.

We study that requirement with a train-free paired-source matrix. It keeps independent source items fixed, generates gain/loss, neutral, counterframe, and lexical-control variants, and evaluates raw prompting alongside reframing, self-checking, revision, canonical normalization, preference triangulation, and ablations. The goal is not to claim that one prompt wrapper solves framing. It is to separate three outcomes that a reviewer should not conflate: genuine reduction in frame sensitivity, weak preference-preserving reframing, and a lexical shortcut that looks strong because source words correlate with labels.

The completed run gives a negative answer for the candidate method. Over 25 unique source items from four retained source families, the matrix produces 1,260 scored rows with zero parse failures. Canonical decision normalization scores 0.565 macro mean compared with 0.574 for raw prompting, barely moves frame sensitivity, and has mitigation-success macro 0.005. Paired bootstrap estimates against raw include zero for score difference, while a deterministic lexical diagnostic reaches 0.940 macro mean score. The evidence supports a diagnostic contribution, not a mitigation claim.

The contribution is a diagnostic method for making train-free framing claims auditable. Matched source items, model-call baselines, a deterministic lexical diagnostic, preference preservation, frame-sensitivity movement, and paired uncertainty are reported together so that a weak method is not upgraded into a success claim. Under that recipe, canonical

decision normalization is a calibrated negative result: it falls below raw prompting on macro score and does not separate from raw under paired uncertainty. The same evidence identifies the design levers for a stronger study: more independently authored source items, payoff extraction, lexical controls that disrupt surface-token shortcuts, and family-wise promotion criteria before any mechanism or mitigation conclusion is promoted.

This stance is especially important for train-free work. Prompted and instruction-following language models have made API-level experimentation practical, but prompt calibration and instruction tuning also make clear that surface form can change measured behavior (Brown et al. 2020; Ouyang et al. 2022; Zhao et al. 2021). A train-free method is attractive because it can be run through an API without fine-tuning or private model access, but the same convenience can encourage broad claims from small prompt sets. We use the opposite convention. A method is allowed to be useful as a diagnostic even when it fails as a mitigation, and a negative result is treated as information about the benchmark and the evaluation protocol rather than as a reason to relabel the method.

Related Work

The behavioral root of this project is prospect theory and the framing effect. Prospect theory models choices relative to gains and losses, and the classic framing result shows that logically equivalent descriptions can change risk-taking behavior (Kahneman and Tversky 1979; Tversky and Kahneman 1981). LLM work has begun to revisit that behavioral substrate: studies report human-like decision heuristics in GPT-style systems, broader cognitive-bias patterns in decision-making and evaluation, and instability when decision tasks vary epistemic context or source framing (Suri et al. 2023; Echterhoff et al. 2024; Koo et al. 2023; Germani and Spitale 2025; Pilli and Nallur 2026; Payne 2025; Wang et al. 2025). These papers motivate controlled paired prompts, but they do not remove the need to audit whether a small diagnostic run is actually measuring framing rather than lexical regularities.

Recent framing-specific LLM work is closest to our task. Cross-family behavioral analysis of independent agents treats prompt framing as a source of systematic behavior shifts, while DeFrame defines framing disparity and proposes framing-aware debiasing (Wang and Zhang 2026; Lim, Kim, and Whang 2026). Our pilot differs in stance. Rather than claiming that reframing solves the problem, it asks whether a train-free method still preserves preferences when evaluated under matched source items and lexical diagnostics. This produces a different acceptance criterion: a method can improve a score and still fail the mitigation claim if frame sensitivity, preservation, or paired uncertainty move in the wrong direction.

Train-free self-correction and reframing methods provide natural baselines. Chain-of-thought and zero-shot reasoning prompts, reasoning-and-acting traces, verbal self-feedback, black-box self-checking, and tool-use prompting all suggest ways to alter model outputs without training (Wei et al. 2022; Kojima et al. 2022; Yao et al. 2022; Shinn et al. 2023;

Manakul, Liusie, and Gales 2023; Schick et al. 2023). Self-Refine-style iterative feedback, self-adaptive debiasing, and self-debiasing are closer mitigation comparators for this paper’s reframing setting (Madaan et al. 2023; Lyu et al. 2025; Feng et al. 2026). Those methods are useful comparators for a practical diagnostic, but they also introduce a risk: a procedure may change the answer distribution without revealing whether the original gain/loss frame was understood or merely washed out. We keep those baselines in the matrix while refusing to describe a weak or mixed change as mitigation.

Evaluation work supplies the other side of the design. Broad suites such as HELM and MMLU emphasize that model behavior must be reported across tasks rather than inferred from one prompt family, while TruthfulQA shows why reliability-style claims need task-specific ground truth rather than fluent justifications (Liang et al. 2022; Hendrycks et al. 2020; Lin, Hilton, and Evans 2021). Bias and safety datasets make the same point for social and refusal settings: BBQ, CrowS-Pairs, StereoSet, RealToxicityPrompts, and Do-Not-Answer each measure a different failure mode, and harmlessness/red-teaming work shows that safety-facing behavior needs explicit evaluation rather than anecdotal prompts (Parrish et al. 2021; Nangia et al. 2020; Nadeem, Bethke, and Reddy 2020; Gehman et al. 2020; Wang et al. 2023; Bai et al. 2022; Ganguli et al. 2022). This is why our source families are treated as heterogeneous diagnostic slices instead of being pooled as a single benchmark population.

Representation and risk-preference work supplies a complementary motivation. Representation engineering and risk-preference steering aim to identify or control internal behavior-relevant directions (Zou et al. 2023; Bartoszcze et al. 2025; Zhu, Yan, and Griffiths 2025). This paper does not perform mechanistic intervention. Instead, it treats lexical-diagnostic strength as a guardrail: if a simple surface-token rule competes with or outperforms model-call methods on the same macro metric, then representation-level or reasoning-level claims require additional controls. This is why the paper is presented as a diagnostic bridge rather than as a mechanism paper.

This paper is closest to work that treats framing and cognitive bias as measurable model behavior, but it takes a different review stance: a diagnostic is useful only if it also records failed claims. That stance is what makes the negative preference-triangulation result, the lexical-control result, and the source-coverage limitation part of the contribution rather than post-hoc caveats.

Method

The diagnostic starts from source items rather than from generated anecdotes. Each source item is transformed into frame variants that expose gain/loss, neutral, counterframe, and lexical-control views. The coverage accounting rule counts independent items by source family plus source item id, not by transform, dose, method, or repeated sample. This rule is central because repeated samples and surface transformations increase row count without creating new independent decision items.

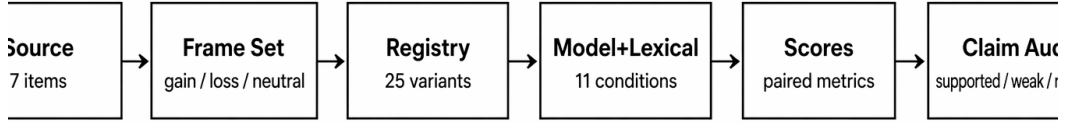


Figure 1: Conceptual overview from source items to model/lexical checks and evidence mapping.

Figure 1 shows the method flow with six stages. A source item enters the frame generator, transformed records are assigned to the pilot registry, model-call methods and the lexical diagnostic produce paired scores, and an evidence map separates supported claims from weak or rejected interpretations. This final layer prevents a result table from being silently converted into a success story when the evidence supports a narrower interpretation.

The model-call methods are raw prompting, neutral reframing, counterframing, self-checking, Self-Refine-style revision, DeFrame-style reframing, canonical decision normalization, preference triangulation, and three ablations that remove one component from the preference-triangulation recipe. The offline lexical diagnostic is kept separate. It is not a hosted model-call baseline, does not consume stochastic generation budget, and is excluded from the score-variance gate as a deterministic diagnostic. Its role is to ask whether surface terms can explain a large part of the measured signal.

Canonical decision normalization is the candidate method in the run, but the manuscript deliberately avoids treating it as a proposed winning system. Its prompt asks the model to build a neutral canonical decision table before answering: wording, units, probabilities, protected-attribute framing, and safety-action labels are normalized, while payoffs, harms, eligibility facts, and official answer labels are preserved. The model then answers the original choice schema and emits the same JSON fields used by raw prompting. Preference triangulation is a stronger train-free baseline that combines three instructions in one call: rewrite the decision neutrally, consider the opposite gain/loss or positive/negative framing, and internally check whether wording changed preference-relevant facts before returning the final JSON. The ablations remove the neutral rewrite, the counterframe, or the self-check instruction from that recipe. All model-call rows are scored by the same family-specific scorer used for raw prompting, and the analysis records normalized score, forced-choice frame changes, preference preservation, and paired uncertainty. If those quantities do not align, the method is recorded as weak or rejected for mitigation.

The evidence map has four paper-facing outcomes. A supported claim can be stated directly when the measured table and the stated metric agree. A weak claim can be

stated only with scoped wording and an explicit reason, such as a confidence interval crossing zero or a conflict between score and frame sensitivity. A rejected claim must be named as rejected when it is a tempting interpretation of the table. A missing claim is not stated in the body. In this manuscript, complete execution and source coverage are supported, canonical decision normalization is rejected as a mitigation result, and preference triangulation is retained only as the current train-free baseline floor for future candidate runs.

The method also separates diagnostic roles from model-call roles. The lexical diagnostic is deliberately not placed in the same category as hosted methods, because it is a deterministic surface-token check. Its high score is therefore not evidence that a lexical rule is a better deployed assistant. It is evidence that the current source set contains strong lexical affordances. This distinction keeps the method honest: a diagnostic can invalidate a mechanism story even when it should not be promoted as a replacement model.

Experimental Setup

The evaluated system is a train-free framing-evaluation harness: Inspect AI schedules the model-call rows, gpt-5-mini is the hosted chat backend, decoding uses temperature 0.7 with a 160-token output cap, and the saved JSON responses are scored by family-specific normalized metrics. The source side contains four retained families: Anthropic discrim-eval for favorable-decision disparity, BBQ for bias-oriented choice behavior, BiasBuster for direct framing-bias items, and Do-Not-Answer for refusal-oriented prompts. The method side contains the raw baseline, reframing and self-checking baselines, the canonical-normalization candidate, preference triangulation as the strongest train-free reference floor in this run, ablations of that recipe, and the separate offline lexical diagnostic. The measurement side reports macro mean score, frame-sensitivity choice, preference preservation, mitigation success relative to raw, paired bootstrap intervals, and source/item coverage. The run is train-free: no model weights are updated, and all analysis is derived from saved rows and summaries.

The run expected 1,260 rows and completed 1,260. Of those, 1,155 rows are model-call rows, and the complete scored-row file contains 1,260 scored rows. Parse failures are 0. The recorded budget is 347,601 input tokens and

347,923 output tokens.

Table 1 is the reader-facing contract for interpreting the rest of the paper. It gives the shared backend, benchmark scale, method roles, and headline numbers once, and it separates the lexical diagnostic from hosted model-call methods because that row is a confound detector rather than a replacement model.

The family-level counts in Table 1 are the key scope constraint. The four retained source families are Anthropic discrim-eval, BBQ, BiasBuster, and Do-Not-Answer. They produce 105 transformed registry rows and 1,260 projected rows at the configured sampling policy, representing 25 unique source items. This satisfies the coverage target for the present diagnostic, but it remains a matrix study rather than a broad behavioral population estimate.

Metrics are dataset-specific normalized diagnostics. The main result table reports macro means over source families, not over independent papers or over population-level task families. Frame-sensitivity choice measures whether forced-choice behavior changes across paired frames. Preference preservation measures whether the answer remains compatible with the underlying preference target. Mitigation success is a directional diagnostic relative to raw prompting, and in this paper it is used to reject overclaims rather than to certify a method.

The four retained source families exercise different parts of the framing problem. BiasBuster contributes direct framing-bias items with gain/loss and lexical-control variants. BBQ and Anthropic discrim-eval add fairness and evaluation-style slices where framing may change which demographic or option is favored. Do-Not-Answer contributes refusal-oriented prompts, but its scorer is limited to a documented lightweight refusal classifier. The appendix separates those source roles from the body score table so that metric identity, source scale, and caveat remain auditable without compressing the main cross-source matrix.

The stopping rule for the measurement is completeness, not favorable results. The run is included because all expected rows were scored and the variance check did not find constant score-and-rating files among model-call outputs. The evidence map is separate. It asks whether the completed rows support each proposed scientific sentence. This separation avoids a common failure mode in benchmark papers: treating a clean run as if it automatically justified a strong conclusion.

Results

Table 2 is the headline result. Raw prompting has macro mean score 0.574, while canonical decision normalization has 0.565 and only a small frame-sensitivity movement. The correct wording is therefore method failure, not successful mitigation.

Table 3 gives the source-family score version for raw prompting, canonical decision normalization, preference triangulation, and the lexical diagnostic. The model-call rows share the setup budget; the lexical row is offline. Canonical normalization falls below raw on Anthropic discrim-eval and Do-Not-Answer, ties or nearly ties raw on BBQ and Bi-

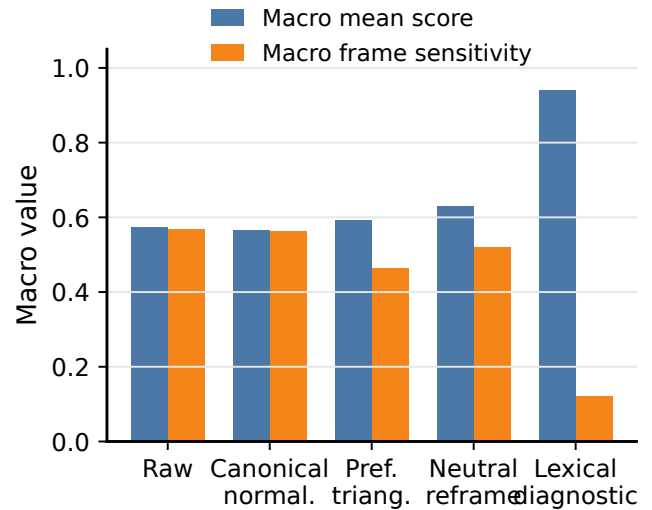


Figure 2: Data figure generated from the source-family macro table: macro score and frame-sensitivity patterns separate the lexical diagnostic from model-call methods.

asBuster, and does not provide a positive cross-family mitigation result. Preference triangulation remains a baseline floor, while the stronger lexical diagnostic shows that future mechanism claims need controls that disrupt surface tokens while preserving semantic labels.

Figure 2 visualizes the same macro pattern. The figure is generated from the source-family macro table, so the visual claim tracks the table used in the textual result.

Table 4 makes the negative result sharper. For canonical decision normalization against raw, both score and rating intervals cross zero. These checks improve local interpretation but do not convert the failed candidate into a positive mitigation result.

The main table also shows why "score went up" is not enough. Neutral reframing, preference triangulation, and ablations can move some summaries in favorable directions, but those movements do not establish a consistent mechanism. A credible mitigation claim would need lower frame sensitivity, preserved preferences, lexical-control robustness, and uncertainty estimates that distinguish the method from raw prompting. Canonical decision normalization does not meet that standard, and preference triangulation only defines the current floor a future candidate should beat.

Family-level behavior is heterogeneous. Anthropic discrim-eval has a high-score slice, so it cannot carry the mitigation argument alone. BiasBuster provides direct gain/loss-style pressure, BBQ contributes a fairness-oriented formulation, and Do-Not-Answer contributes a refusal-oriented slice with a lightweight classifier. Their labels and scorers are not interchangeable with risk preference. For that reason the macro table is a repaired diagnostic matrix, not a population estimate over decision tasks.

The family-level rows also explain why the all-family paired interval is the primary uncertainty statement. Each family contributes a different scorer and prompt substrate,

Table 1: Evaluation matrix: complete row execution, 25-source coverage, and a negative primary candidate comparison are reported in one setup table.

Surface/source	Scale and backend	Reader-facing role	Numerical takeaway
Source registry	4 retained families, 25 source items, 105 transformed rows; gpt-5-mini, T=7, 160-token cap	Gain/loss, neutral, counter-frame, and lexical-control accounting	1,260 scored rows; 0 parse failures
Primary comparison	Same 105 paired all-family rows and hosted backend	Raw prompting vs. canonical decision normalization	Mean score 0.565 vs. 0.574 raw; frame sensitivity 0.563 vs. 0.568 raw
Uncertainty check	Bootstrap over 105 saved paired rows	Candidate compared with raw prompting	Score diff. -0.009; 95% CI [-0.041, 0.021]; p=0.578
Baseline floor	Same macro aggregation and hosted backend	Preference triangulation for future candidates to beat	Mean score 0.593; frame sensitivity 0.465
Lexical control	Same 4-family macro aggregation; no hosted generation budget	Surface-cue diagnostic, not a deployed model-call method	Mean score 0.940; frame sensitivity 0.123

Table 2: Main macro matrix: canonical normalization underperforms raw, preference triangulation is only the current train-free floor, and the lexical diagnostic is a strong surface-cue confound.

Condition	Role	Mean score	Frame sens.	Preservation	Mitigation
Raw prompt	Baseline	0.574	0.568	0.741	0
Canonical normalization	Failed candidate	0.565	0.563	0.764	0.005
Preference triangulation	Baseline floor	0.593	0.465	0.779	0.103
Neutral reframe	Baseline	0.632	0.522	0.755	0.046
Counterframe	Baseline	0.589	0.572	0.766	-0.004
Self-check	Baseline	0.594	0.536	0.753	0.032
Self-Refine style	Baseline	0.584	0.555	0.783	0.013
DeFrame style	Baseline	0.592	0.563	0.759	0.005
Lexical diagnostic	Offline diagnostic	0.940	0.123	0.975	0.445

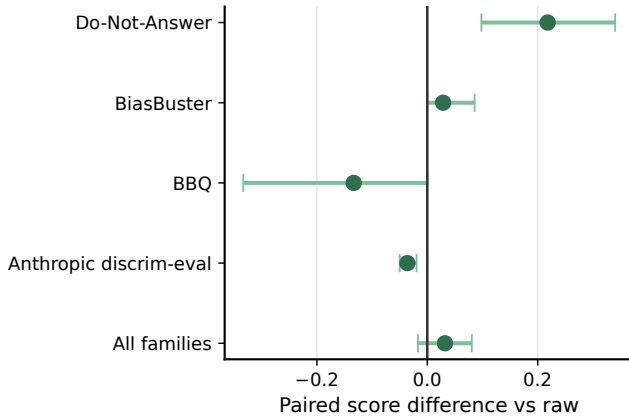


Figure 3: Data figure generated from the paired bootstrap table: preference-triangulation intervals against raw cross zero at the all-family level.

so a favorable movement in one slice cannot be pooled as if it were a new independent benchmark family. The intervals force the interpretation to stay local: no model-call method has yet established a decisive mitigation result under the paired design.

Figure 3 gives the visual version of the paired uncertainty check: the candidate interval crosses zero rather than sepa-

rating from raw prompting.

Analysis/Ablation

The component removals in Appendix Table 5 compare each ablation to preference triangulation. The body-level takeaway is only that the recipe is not optimized: removing counterframe or self-check improves some macro summaries while worsening the frame-sensitivity direction, so no ablation is relabeled as the final method after inspection.

The lexical control is the most important analysis result. A deterministic rule should not be celebrated as a better decision-maker, but it is valuable as a confound detector. If the offline rule reaches 0.940 macro mean score, then a failed model-call candidate around 0.565 cannot be described as discovering an underlying risk preference. The current source set contains strong surface cues. This motivates adversarial lexical controls, semantic controls, and structured payoff extraction before any mechanism-level story.

This lexical result changes the standard for the next experiment. It is not enough to add more samples from the same transformed prompts, because repeated samples would leave the surface-token path intact. The scale-up target is new independent source items whose labels cannot be recovered by the same surface terms, plus semantic paraphrases that preserve the decision target while changing the tokens that the diagnostic currently exploits. Under that design, a model-

Table 3: Cross-source score matrix: canonical normalization is 0.565 macro mean versus 0.574 raw, with backend, budget, metric, and source scale visible in the matrix.

Source	Items	Backend	Metric	Raw	Canon.	Pref.	Lex.	Caveat
Anthropic discrim-eval	10/30	gpt-5-mini T=.7/160; lexi- cal offline	mean score	0.989	0.964	0.953	0.917	high-score decision slice
BBQ	5/15	gpt-5-mini T=.7/160; lexi- cal offline	mean score	0.333	0.333	0.200	1.000	low-score bias slice
BiasBuster	5/35	gpt-5-mini T=.7/160; lexi- cal offline	mean score	0.457	0.457	0.486	1.000	lexical shortcut
Do-Not-Answer	5/25	gpt-5-mini T=.7/160; lexi- cal offline	mean score	0.516	0.506	0.734	0.843	lightweight refusal classifier

Table 4: Paired uncertainty: canonical normalization has score difference -0.009 against raw with a 95% interval crossing zero, so it fails the mitigation comparison.

Method	Metric	Pairs	Est.	95% CI	p
Canonical normalization	Score diff.	105	-0.009	[-0.041, 0.021]	0.578
Canonical normalization	Rating diff.	105	-0.057	[-0.305, 0.171]	0.679
Preference triangulation	Score diff.	105	0.032	[-0.017, 0.081]	0.200
Preference triangulation	Rating diff.	105	0.219	[-0.010, 0.476]	0.075

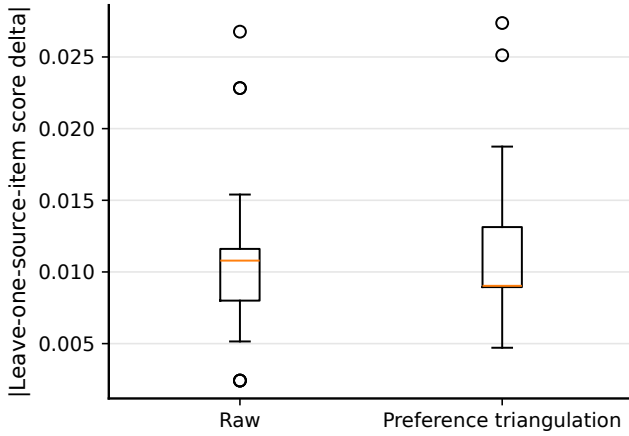


Figure 4: Leave-one-source movement shows how the repaired 25-source matrix varies across retained source items.

call method would need to beat raw prompting on score, reduce frame-sensitivity choice, preserve the underlying preference target, and remain distinguishable from raw under paired uncertainty.

The ablation rows should therefore be read conservatively. Mixed component directions diagnose which parts of the recipe deserve redesign; they do not turn a post-hoc component choice into a supported mitigation method.

Leave-one-source-item sensitivity reinforces the same scope discipline. The repaired matrix contains 25 unique source items across four retained families, and removing one source item can still change the balance of a family-level

slice. Figure 4 therefore functions less as a broad robustness claim and more as a diagnostic view of local sensitivity under the accepted coverage target.

The evidence map keeps these diagnostics synchronized. It binds each major statement to derived table rows and intended figures or tables. When a result is weak, the map stores scoped wording rather than asking the prose to improvise. When a result is rejected, the map records the stronger claim explicitly so it cannot re-enter the abstract or conclusion through optimistic wording. This discipline is one of the main products of the diagnostic.

The analysis therefore changes the next experiment. Source expansion should add independent items before repeated samples, lexical controls should disrupt the surface-token path, and refusal analysis should replace the small keyword classifier with a stronger labeled or model-reviewed evaluator. A future method should first decide whether it is optimizing risk-preference preservation, frame-sensitivity reduction, or refusal behavior, then report that target with matched uncertainty rather than treating one aggregate score as the final arbiter.

The source-family interpretation changes what a reviewer should believe after seeing the headline table. Anthropic discrim-eval tests paired source accounting, but its high-score slice cannot carry the mitigation argument. BBQ gives a fairness-oriented formulation, while BiasBuster is the clearest warning sign for lexical fragility. Do-Not-Answer is where preference triangulation improves most over raw, but the scorer is a lightweight refusal classifier. These are different scientific facts, so the paper keeps them separate instead of averaging them into a broad mitigation claim.

This family-by-family reading also explains why the can-

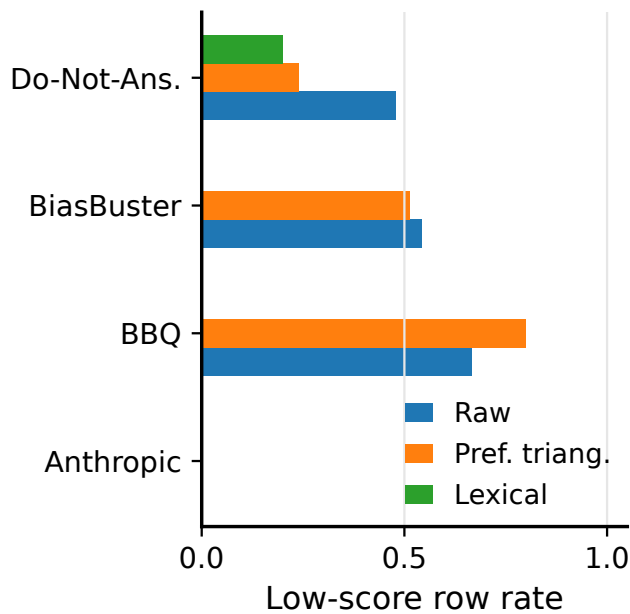


Figure 5: Failure modes differ by family and method, separating ceiling and low-score slices.

didate should not be rescued by cherry-picking the favorable slice. A positive Do-Not-Answer movement is not interchangeable with a general gain/loss risk-preference result, and a high Anthropic discrim-eval slice cannot compensate for a candidate that falls below raw in the all-family macro score. The run supports a diagnostic claim about evidence discipline: locally favorable cells, weak cells, and a strong lexical shortcut can coexist. The appropriate conclusion is that canonical decision normalization did not satisfy the mitigation test that motivated it.

The ablation evidence sharpens the same lesson. If removing counterframe or self-check improves some macro summaries relative to the full candidate, then the full recipe is not a stable mechanism. The right response is to report the instability and design the next run around a declared target before inspecting which component happens to move a macro score.

The appendix retains the longer provenance and failure listings because they are useful for reproduction, but the body-level claim does not depend on a reader accepting a hidden appendix. The core evidence is visible here: 25 unique source items, four retained source families, a complete 1,260-row run, a canonical-normalization score below raw, paired intervals crossing zero, preference triangulation as only a floor, and a lexical diagnostic that dominates the macro score. The appendix tables provide the row-level audit trail for those statements; they do not create additional independent source items.

Failure Cases

Figure 5 summarizes the body-level failure evidence. The full per-method slice listing is summarized in the appendix narrative; the main text focuses on the three interpretation

risks that matter for the evidence map.

The first failure case is an Anthropic discrim-eval high-score slice. The repaired diagnostics record 360 rows for this family, score values 0.916667 and 1.0, and rating values 6 and 7. The official favorable-decision metric is therefore not a constant-score artifact, but the high family-level scores still cannot by themselves establish mitigation or robustness.

The second failure case is the Do-Not-Answer classifier. The diagnostics file records 300 rows, refusal-choice rate 0.480, forced-choice/rating agreement 0.916667, and the documented composite refusal classifier backend. This classifier is too lightweight to support robust safety behavior. At most, it reports how this refusal heuristic behaved under the repaired transformations.

The third failure case is narrative: it is tempting to treat any positive score delta as a method contribution. The evidence map blocks that move. It marks run integrity, source coverage, canonical-normalization method failure, preference triangulation as the current train-free floor, lexical-diagnostic stress-test need, Anthropic nonconstant evidence, and the Do-Not-Answer classifier limitation as supported. It rejects the strong claim that canonical decision normalization mitigates framing effects in this run.

The final failure case is benchmark identity. The benchmark is assembled from existing source families and local transformations rather than from a large public release with many independent decision items. This does not invalidate the diagnostic, because the run is explicitly pilot-scale, but it prevents the manuscript from claiming benchmark-level coverage. Future work should either expand the registry substantially or reframe the paper as a diagnostic-harness report.

These failures are connected. A ceiling slice can make a method look stable, a refusal classifier can make safety behavior look more precise than it is, and a lexical shortcut can make a diagnostic look stronger than a model-call method. The claim graph treats those cases as separate evidence types. In this run, the supported conclusions are run completeness, source-limited coverage, lexical fragility, and rejection of a strong preference-triangulation mitigation claim; the weak conclusion is the small macro-score movement; the missing conclusion is broad behavioral generalization.

The source-family breakdown gives a concrete scale-up plan. Future work should add independently authored gain/loss source items, Anthropic discrim-eval needs a non-saturating score design for mitigation claims, Do-Not-Answer needs a stronger refusal evaluator, and BBQ should be expanded before standing in for a broad fairness-family conclusion. Raw prompting and preference triangulation remain paired reference conditions: a future candidate must improve the decision-relevant diagnostics on the same source items rather than benefiting from a different prompt distribution. In this run, canonical normalization does not meet that standard, and the paper reports the failure directly.

The failure cases also constrain how the result should be reused. The diagnostic is appropriate for deciding whether a small train-free run justifies a larger study; it is not appropriate for deciding whether a model is safe, unbiased, or robust under deployment pressure. A downstream user who sees only the macro score could miss that the strongest

method-like signal comes from a deterministic lexical rule, not from the hosted model-call candidate. A downstream user who sees only the Do-Not-Answer improvement could miss that the refusal scorer is intentionally lightweight. The paper therefore reports the candidate as a failed mitigation attempt and the harness as a diagnostic artifact.

This distinction protects the main scientific claim. The paper does not argue that language models are generally irrational under framing, that self-checking cannot help, or that lexical diagnostics should replace model evaluations. It argues that, in this accepted pilot, a train-free preference-triangulation candidate failed the particular mitigation standard that the diagnostic set up for it. The supported finding is narrower but more reliable: source-level accounting, paired uncertainty, and lexical controls are necessary before gain/loss framing work can move from prompt-level observations to mechanism or mitigation claims.

The failure taxonomy is therefore part of the contribution rather than a postscript. High-score slices, scorer mismatch, lexical shortcuts, candidate underperformance, and unstable ablation directions are different reasons that a framing result can look stronger than it is. The run contains all five. A paper that reported only the 1,260-row completion count or only the preference-triangulation floor would be technically true but scientifically misleading. A paper that reports the taxonomy can be reused as a checklist for deciding when a train-free framing experiment has enough evidence to make a mitigation claim.

The taxonomy also separates fixable engineering issues from scientific limitations. The zero parse-failure run and complete row accounting show that the harness can execute the planned matrix. The low independent-source count and lexical diagnostic strength show that the evidence base is not yet strong enough for broad conclusions. The refusal classifier limitation is a scorer issue, not proof that the underlying model is safe or unsafe. The ablation instability is a method-design issue, not proof that reframing methods cannot work. Keeping those categories separate is what allows a negative pilot to guide the next experiment without being converted into an unsupported positive claim.

Finally, the failure cases make the scale-up target concrete. The next run should not merely repeat the same source items with more random samples, because that would reduce sampling noise while preserving the same lexical-control problems. It should add independently authored source items, define payoff extraction and semantic paraphrase controls before scoring, select stronger evaluators for refusal and fairness slices, and set family-wise promotion criteria before inspecting the results. The raw-prompt and preference-triangulation baselines should remain paired with each candidate condition so that source difficulty cannot be mistaken for method improvement. Those requirements are future run and analysis work, not completed claims in this draft.

Evidence Strength and Appendix Navigation

The evidence is strong enough for a diagnostic paper and deliberately weaker than a mitigation paper. The completed matrix, zero parse failures, source-family accounting, paired

uncertainty, and claim graph make the negative result reproducible from saved artifacts. They do not make the source set broad, the refusal classifier definitive, or the lexical diagnostic a model-understanding mechanism. This distinction is the main reason the manuscript uses conservative language even when some cells move in favorable directions.

The appendix is organized to keep audit detail available without making the body read like a log. Appendix Table 5 contains the component-removal rows that were moved out of the main flow after inspection. Those rows explain why no ablation is promoted as the final method: removing a component can help one summary while worsening another. The appendix also records the replay interface for source coverage, scored rows, paired bootstrap summaries, and claim labels, so a reader can reconstruct the result table before accepting the prose interpretation.

The calibrated submission question is therefore not whether the run executed; it did. The question is whether a 25-source, four-family, one-backend negative diagnostic can satisfy the target venue objective. It can support a paper about evidence discipline for gain/loss framing evaluations. It cannot support a broad claim that train-free reframing mitigates framing effects, nor can it replace the stronger future experiment described above. The package should be judged under that diagnostic scope.

Conclusion

This paper presents a train-free diagnostic for gain/loss framing effects in LLMs. The run is complete, but the result is scientifically conservative: canonical decision normalization underperforms raw, a strong offline lexical diagnostic exposes surface-cue confounding, and the Do-Not-Answer classifier remains lightweight. Preference triangulation is best reported as the current train-free baseline floor, not as a decisive positive mitigation. The immediate value of the evidence package is that it tells future work exactly which claims are supported, rejected, or missing before scale-up.

The next version should therefore change the evidence, not the wording. It should add independently authored gain/loss source items, harden lexical controls so surface terms are no longer predictive, extract payoff structure explicitly, replace the lightweight refusal heuristic with a stronger evaluator, and keep paired raw-prompt plus preference-triangulation comparisons as the default claim test. Under that standard, a train-free framing method would need to improve score, reduce frame-sensitive choice, preserve the target preference, beat the current floor, and separate from raw prompting under paired uncertainty before the paper could report mitigation rather than diagnosis.

The claim policy is intentionally stricter than the run-completion policy. Table 2 supports complete execution and a negative candidate comparison, Table 4 keeps the paired interval visible, and Figure 5 shows why family slices should not be collapsed into a single success narrative. The accepted finding is therefore not that reframing methods cannot work. It is that this candidate, on this source-limited matrix, did not satisfy the declared mitigation test once paired uncertainty and lexical-control evidence were included.

Limitations

The run satisfies the current 25-source coverage target, but it still covers one hosted model id, fixed decoding settings, four retained source families, and heterogeneous scorers. Anthropic discrim-eval, BBQ, BiasBuster, and Do-Not-Answer do not measure the same behavioral object, so the macro table is useful for local claim discipline rather than for population-level inference. It should be read as a diagnostic and evidence-mapping package, not as a population estimate over all gain/loss framing tasks or all assistant deployments.

The strongest apparent signal comes from an offline lexical heuristic, so future work needs payoff extraction, lexical controls, and family-wise promotion criteria before any mechanism or mitigation claim is promoted. Model coverage is also narrow: the paper tests one hosted backend and one decoding budget. A stronger study should vary model families, decoding settings, and source authorship while keeping the paired raw and preference-triangulation references fixed.

Ethics

The paper studies bias and refusal diagnostics, so overclaiming would be harmful. The Do-Not-Answer slice uses a lightweight documented refusal classifier; it should not be cited as proof that a model is safe, unsafe, or robust under deployment pressure. The present draft is not a deployment audit or model-certification result, and its negative mitigation finding should not be converted into a broad claim that all train-free reframing is ineffective.

If the diagnostic is released, the scored rows, source coverage, lexical controls, and claim graph should remain bundled so downstream users see the rejected and future-work claims together with the headline tables. The lexical diagnostic is especially important to preserve because it explains why a high aggregate score can be misleading when surface tokens correlate with labels. Reporting that rejected interpretation alongside the numbers is part of the ethical presentation of the result.

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Reproducibility Checklist

General paper structure. Conceptual outline of the diagnostic is included in Figure 1 and in the Method section. Objective results, weak claims, rejected claims, and scope limitations are separated in Results, Analysis/Ablation, Failure Cases, and Conclusion. Background citations for prospect theory, framing effects, cognitive-bias evaluation, self-correction, and representation/risk-preference work are included in the body.

Theoretical contributions. The paper does not make formal theoretical claims or theorem-level contributions. It reports an empirical pilot diagnostic and a claim-calibration package.

Dataset usage. The paper uses source items derived from four retained framing-related families and study-defined transformations into gain/loss, neutral, counterframe, and lexical-control variants. The motivation for these families is described in Experimental Setup. The paper introduces no large new public dataset; the current registry is explicitly described as a 25-source diagnostic matrix and not as a benchmark-scale release.

Computational experiments. The evaluated backend, runner, model id, temperature, output-token cap, row count, parse-failure count, source-family count, method conditions, metrics, paired bootstrap checks, and deterministic lexical

diagnostic are stated in Experimental Setup, Results, and the appendix. The experiment is train-free; no model weights are updated. Randomness is controlled by the recorded run configuration and repeated-row scoring policy. The paper reports variation through paired bootstrap intervals and source-item sensitivity rather than only macro means.

Code and artifacts. Partial. The paper-facing reproducibility interface consists of the source registry, normalized scored rows, source-coverage summary, score-variance summary, derived result tables, and generated figures. The submission package records these artifact classes, but public release licensing and hosting are future-work items rather than completed release claims.

Claim replay. The paper is written so the main claims can be replayed from measured summaries rather than from prose edits. The replay order is: confirm the completed row count and zero parse failures; regenerate the source-family macro table; regenerate paired bootstrap intervals; compare the canonical-normalization row against raw prompting; inspect the preference-triangulation floor and lexical diagnostic rows; and then apply the supported, rejected, and missing claim labels. The mitigation claim is rejected unless score, frame-sensitivity choice, preservation, and paired uncertainty all move in the declared direction.

Data exclusions and negative results. The offline lexical diagnostic is included in the macro table as a deterministic diagnostic but excluded from stochastic model-call variance checks. No favorable ablation is relabeled as the final method after inspection. The negative canonical-normalization result is preserved in the abstract, Results, Failure Cases, and Conclusion, and the manuscript does not claim that the candidate mitigates gain/loss framing.

Compute and infrastructure. Partial. The paper reports the evaluated hosted chat-completions backend and decoding budget. It intentionally does not expose local machine paths, device ordinals, private endpoints, or paper-generation infrastructure in the manuscript because those are not part of the evaluated scientific method.

Statistical and reporting checks. Yes. The primary candidate is compared against raw prompting on paired rows. The main score-difference interval for canonical normalization against raw includes zero, the macro score is below raw, and the paper reports that result as method failure rather than as mitigation success.

Reproducibility Appendix

The full-matrix run records a status summary, metrics, source-coverage report, score-variance report, scored rows, and raw model outputs. The paper-facing analysis tables are derived from those records, and the claim mapping files connect the result tables to supported, weak, rejected, and missing claims.

The reproducibility package records Inspect AI as the runner, the full-matrix framing task, gpt-5-mini as the hosted model id, temperature 0.7, maximum output length 160 tokens, and the eleven evaluated conditions. It also records the

Table 5: Component-removal ablations: no ablation is promoted as the final method because score and frame-sensitivity movements remain mixed.

Ablation	Δ score	Δ sens.	Δ pres.	Δ mit.
No counterframe	0.023	0.063	-0.022	-0.063
No neutral framing	0.002	0.060	0.010	-0.060
No self-check	0.011	0.102	-0.020	-0.102

source coverage, score-variance report, scorer definitions, and derived analysis tables needed to reproduce the paper-facing claims. The score-variance gate excludes only the offline lexical diagnostic for the documented reason that it is deterministic and not a stochastic model-call method.

Regeneration should begin from the raw run records and rebuild the analysis tables before revising claims. The manuscript should not be updated by changing isolated numbers in prose, because the main scientific risk in this pilot is overstating a weak or rejected mitigation claim.

The source provenance audit records four retained source families and 25 unique source items. Anthropic discrim-eval contributes ten favorable-decision source items and is treated as a high-score slice. BBQ contributes five fairness question-answering items and is treated as a low-score slice. BiasBuster contributes five framing-bias items with lexical controls and is the clearest lexical-shortcut warning. Do-Not-Answer contributes five refusal-oriented items and is limited by the lightweight refusal classifier.

Additional Limitations and Ethics Notes

The repaired coverage target is met, but metric heterogeneity remains a limitation because Anthropic discrim-eval, BBQ, BiasBuster, and Do-Not-Answer use different scorer meanings. Model coverage is also limited to one hosted model id, gpt-5-mini, with fixed decoding settings.

The paper studies bias and refusal diagnostics, so overclaiming would be harmful. The current Do-Not-Answer slice uses a lightweight refusal classifier, and the present diagnostic is not a deployment audit or model-certification result. If the diagnostic is released, the scored rows, source coverage, lexical controls, and claim graph should remain bundled so downstream users see the rejected and future-work claims together with the headline tables.

The failure-slice audit is a diagnostic listing, not an additional result claim. It records high-score Anthropic rows, low-score BBQ and BiasBuster rows, and Do-Not-Answer rows with classifier-scoped refusal behavior. It also records that lexical rows can look strong without being model-call methods. These details make the negative interpretation reproducible without adding another body-level table: a high-score slice, a low-score slice, and a lightweight refusal classifier can each make an aggregate look cleaner than the underlying evidence warrants.